Data-Efficient Dense Self-Supervised Learning for Breast Tumour Segmentation

1 Context of the internship

The internship will take place at Télécom Paris, within the Image, Données, Signal (IDS) Department, and more specifically in the IMAGES research team. The internship is expected to start in spring 2027 and will have a duration of four to six months. It will be conducted in a research-oriented environment, with close interaction with academic researchers and PhD students working on medical imaging and machine learning.

2 Internship issue

Breast tumour segmentation supports diagnosis, treatment planning and response monitoring, but supervised models require costly expert voxel-level annotations and generalise poorly across centres, scanners and protocols. Dense self-supervised learning (SSL) learns local representations without labels, so that only a few annotated cases are needed downstream. This internship investigates how dense SSL can be adapted to breast tumour segmentation, a setting far from the anatomical tasks in which these methods were developed.

3 Analysis of the current situation

Dense SSL methods, such as vox2vec [1], Pix2Rep [2] and Pix2Rep-v2 [3], have demonstrated strong capabilities for the segmentation of anatomical structures (e.g. cardiac MRI, abdominal CT). Their main advantages are data-efficiency, with competitive performance from only a few annotated scans, scalability to 3D, and, for Pix2Rep-v2, training-free in-context segmentation. However, these structures are large and stable in appearance across subjects. Breast tumours are instead small, highly variable pathological targets surrounded by confounding tissue. Whether dense SSL can learn tumour-level semantics, therefore remains open. The MAMA-MIA benchmark [4], with 1,506 Multi-centre annotated cases and external evaluation are an ideal testbed to address this question.

4 Scientific proposal

The student will develop a dense SSL method capturing tumour-level semantics, building on Pix2Rep-v2 [3] and starting from MAMA-MIA, and will compare it against the state-of-the-art segmentation methods. The project will focus on:

- Designing tumour-aware dense SSL pre-training;
- Benchmarking against SOTA segmentation methods, including nnU-Net [5] and foundation models, from one-shot to full-data regimes;
- Assessing robustness across centres and the quality of the learned tumour representations.

5 Expected results

The internship will deliver an adapted dense SSL pipeline for breast tumour segmentation, evaluated on the MAMA-MIA Codabench benchmark. Results are expected to show that dense SSL substantially reduces annotation needs and improves cross-centre generalisation, paving the way towards label-efficient breast cancer imaging models. Potential extensions include other breast cancer datasets and imaging settings, and other use cases leveraging the learned tumour representations, such as prediction of response to neoadjuvant chemotherapy or longitudinal tumour monitoring.

How to apply?

- Contact: `maxime.difolco@telecom-paris.fr` and `loic.lefolgoc@telecom-paris.fr`
- Documents to include in the application: resume, transcripts.

[1] Mikhail Goncharov et al. "vox2vec: A Framework for Self-supervised Contrastive Learning of Voxel-Level Representations in Medical Images". In: *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2023, pp. 605–614.

[2] Maxime Seince et al. "Dense Self-Supervised Learning for Medical Image Segmentation". In: *Medical Imaging with Deep Learning (MIDL)*. 2024.

[3] Sofiane Sifaoui et al. "Pix2Rep-v2: Data-Efficient Representation Learning for Dense Medical Imaging Applications". In: *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. arXiv:2609.01427. 2026.

[4] Lidia Garrucho et al. "A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations". In: *Scientific Data* 12.1 (2025), p. 453. DOI: 10.1038/s41597-025-04707-4.

[5] Fabian Isensee et al. "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation". In: *Nature Methods* 18.2 (2021), pp. 203–211.